ABSTRACT

QuantumSHIELD is a security framework designed to strengthen enterprise IT environments against evolving cyber threats and future quantum computing risks. The proposed system integrates AI-based security incident analysis , post-quantum cryptographic protection , permissioned blockchain based audit mechanisms, and controlled incident response into a unified security workflow. Kubernetes security observations are processed to identify and correlate abnormal conditions, after which the AI component analyzes incidents and provides classification, severity, and recommended actions. SHA-256 hashing is used to maintain the integrity of security records, while ML-DSA and ML-KEM interfaces provide an architecture for future post quantum digital signatures and key encapsulation. A blockchain audit layer provides tamper-evident recording of security events , while role-based access control and human approval mechanisms prevent unauthorized remediation actions. The system was functionally validated through a local demonstration environment using deterministic Kubernetes observations, AI analysis, mock blockchain recording, simulated response, and audit verification. The results demonstrate an integrated workflow for security monitoring, incident analysis, integrity protection, controlled response, and auditable security operations